

Enhancing IEEE 802.11 Standard with Deep Reinforcement Learning for Optimal Channel Access

Sheila Cássia da Silva Janota Cruz

sheila.cassia@mtel.inatel.br

Advisor: Felipe Augusto Pereira de Figueiredo

Instituto Nacional de Telecomunicações - INATEL

Master's degree in Telecommunications
Santa Rita do Sapucaí, June 24th of 2024

Outline

❖ Introduction

- Motivation
- Problem Statement
- Research Objectives
- Publications

❖ Machine Learning Overview

- Reinforcement Learning
- Deep Reinforcement Learning
- Multi-Agent Reinforcement Learning

❖ Methodology

- Network Topology: centralized and decentralized solution
- Observation Metrics
- Action Space
- Reward
- Simulation Setup

❖ Results and Analysis

- Centralized Results
- Centralized vs Decentralized results

❖ Conclusions and Future Work

Motivation

- The carrier-sensing multiple access with collision avoidance (CSMA/CA) mechanism adopted by the IEEE 802.11 standard uses a binary exponential backoff (BEB) algorithm to prevent collisions and improve channel access.
- The BEB algorithm presents some drawbacks:
 - Inefficiency under high loads;
 - Inability to adapt to changing networks;
 - Lack of fairness among nodes;
 - Vulnerability to the hidden node problem;
 - No global optimization.
- Therefore, the application of the BEB algorithm in dense networks causes bandwidth wastage and reduces the network's throughput dramatically, leading to inefficient network operation.

Problem Statement

- The BEB is not the most efficient solution, leading to various challenges and performance issues:
 - Bandwidth wastage;
 - Decreased network performance in dense networks;
 - Incorrect backoff values increase collision occurrences drastically, making channel access extremely challenging;
 - Scalability problems as the stations in the network increases.

- These limitations and drawbacks highlight the need for an optimized solution to enhance the collision avoidance mechanism in IEEE 802.11 networks.

Main Research Objective

- Employing reinforcement learning (RL) algorithms to:
 - Improve network performance;
 - Reduce collision occurrences;
 - Enhance scalability in dense and dynamic scenarios.

- By applying RL algorithms to the collision avoidance problem, we aim to find an optimized solution that outperforms the BEB algorithm.

Publications

Publication in International Journal

1. **Sheila Cássia da Silva Janota Cruz**, Messaoud Ahmed Ouameur, and Felipe Augusto Pereira de Figueiredo, "Reinforcement Learning-based Wi-Fi Contention Window Optimization," Journal of Communication and Information Systems, vol. 38, no. 1, pp. 128–143, 2023. DOI: 10.14209/jcis.2023.15.

Publication in International Conference

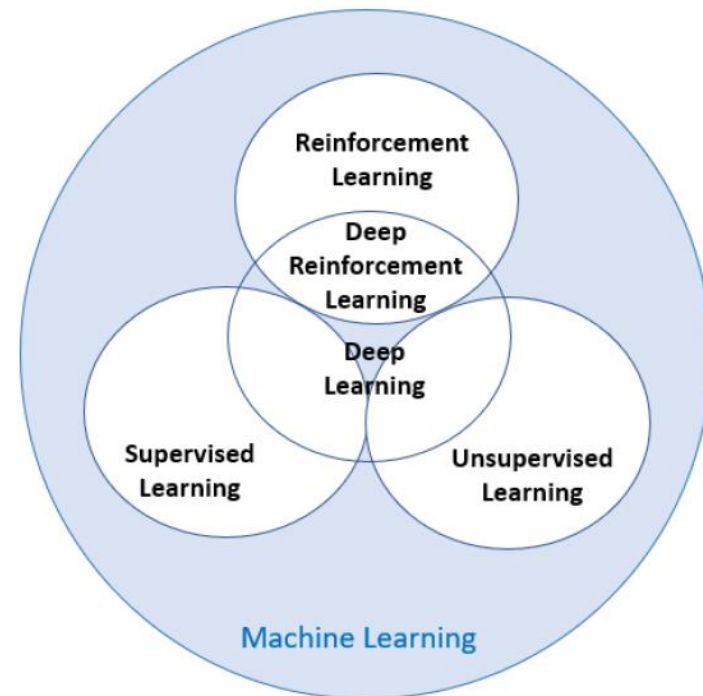
2. **Sheila Cássia Da Silva Janota Cruz**, Felipe Augusto Pereira de Figueiredo and Messaoud Ahmed Ouameur, "Enhancing IEEE 802.11 Standard with Deep Reinforcement Learning for Optimal Channel Access, "2023 3rd International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)", Tenerife, Canary Islands, Spain, 2023, pp. 1-6, DOI: 10.1109/ICECCME57830.2023.10252258.

Submitted in National Conference

3. **Sheila Cássia Da Silva Janota Cruz**, Felipe Augusto Pereira de Figueiredo and Rausley Adriano Amaral de Souza, "Decentralized Deep Reinforcement Learning Approach for Channel Access Optimization," 2024 XLII Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT), Belém, Pará, Brazil.

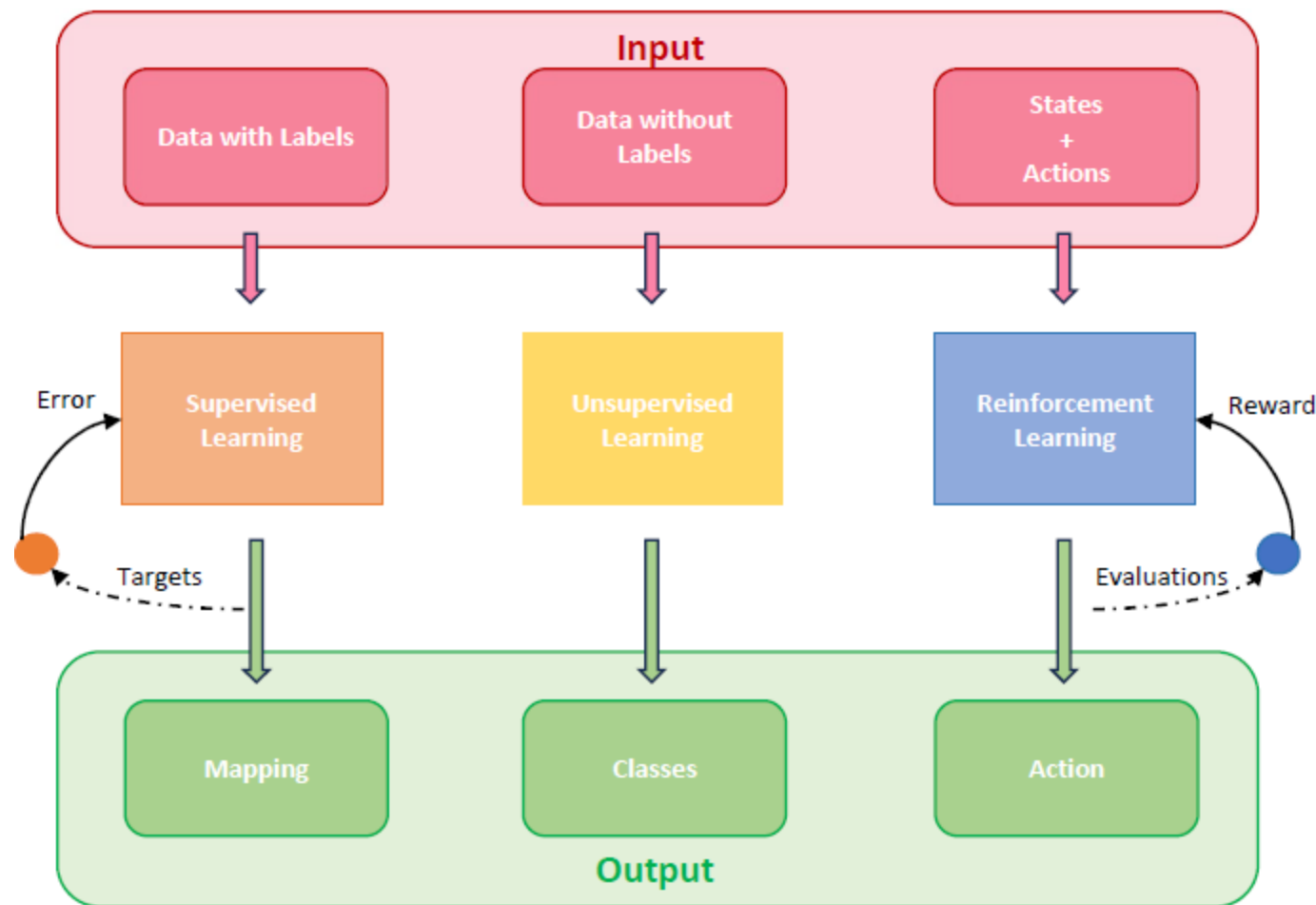
Machine Learning Overview

- ML is a set of intelligent algorithms (model) that learn characteristics of a system that can not be explicitly presented by a mathematical expression.
- ML models are trained based on historical data to perform tasks.
- The learning process of an ML model is called training, and it is used for the model to gain knowledge and achieve the desired result.
- Based on the type of training the learning process can be classified in three paradigms: **Supervised learning**, **Unsupervised learning** and **Reinforcement Learning (RL)**.
- Provide self-management, self-learning, and self-optimizing solutions.



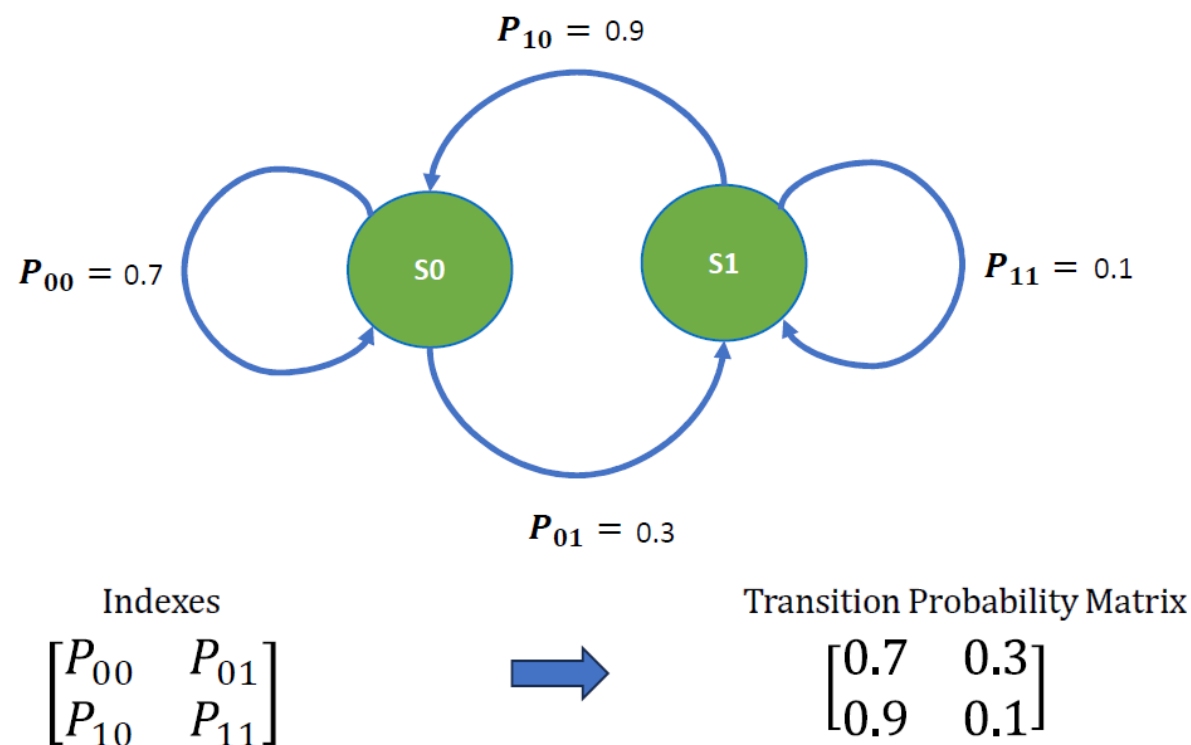
Machine Learning Overview

➤ Learning Paradigms



Reinforcement Learning Theoretical Background

➤ Markov Chain State Transition



Reinforcement Learning Theoretical Background

➤ Markov Decision Process

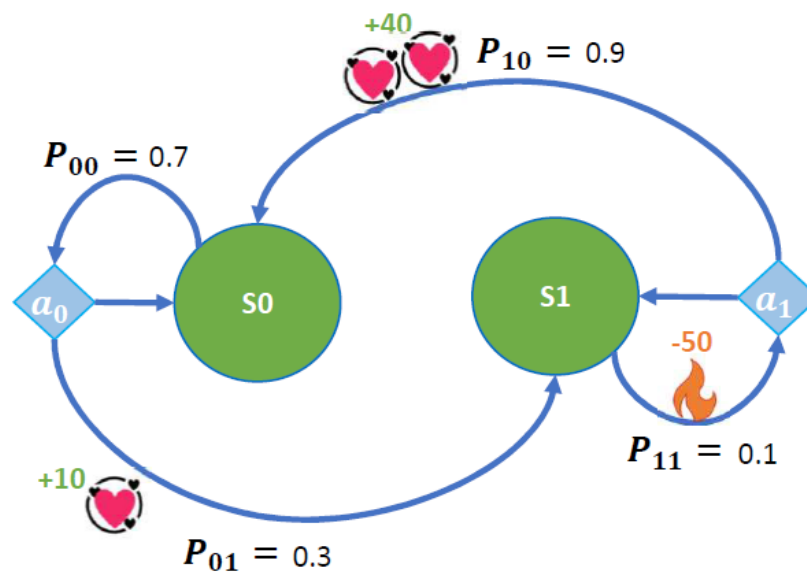


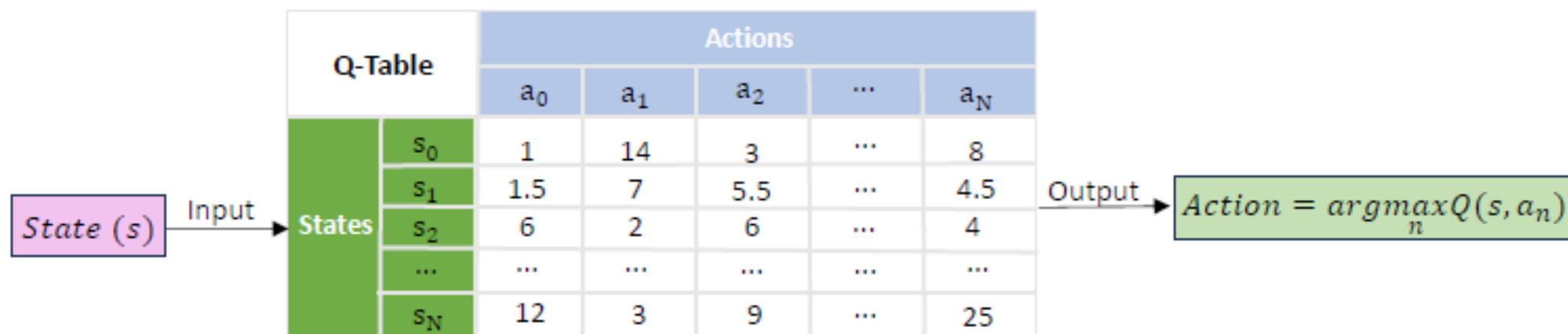
Figure 2.3: Markov Decision Process State Transition

Indexes		Actions		Rewards	
Action (Columns)		Action (Columns)		Action Rewards (Columns)	
State (Rows)		a_0	a_1	a_0	a_1
s_0	$\begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix}$	$\begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix}$	$\begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix}$	$\begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix}$	$\begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix}$
s_1	$\begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix}$	$\begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix}$	$\begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix}$	$\begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix}$	$\begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix}$

Figure 2.4: Markov Decision Process State Transition Matrix

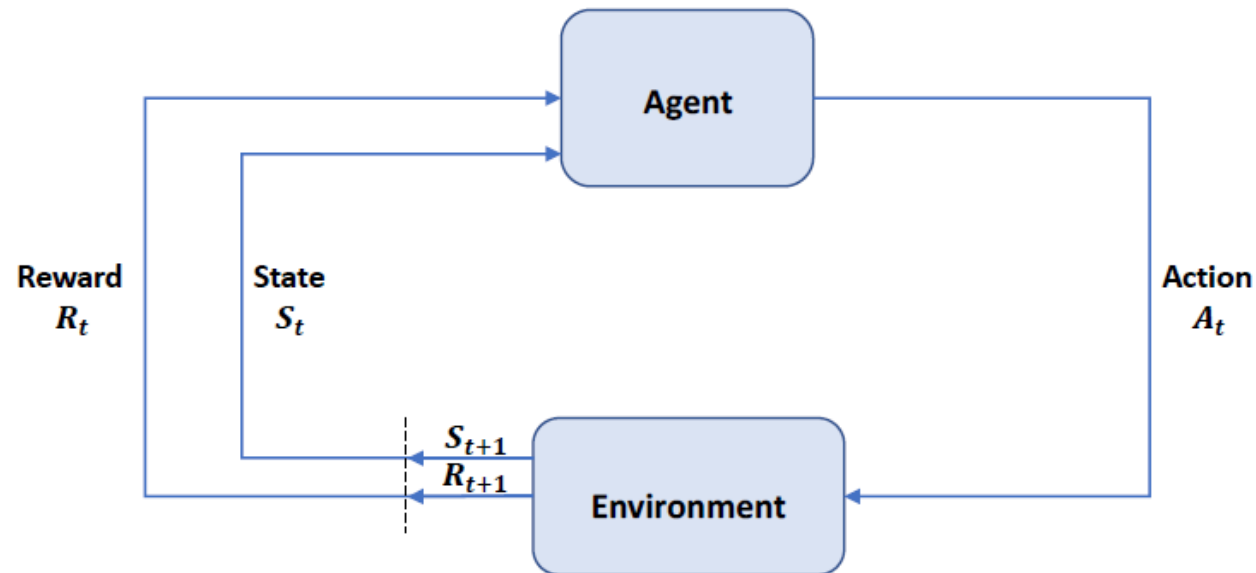
Reinforcement Learning Theoretical Background

➤ Basic idea of Simple Q Learning



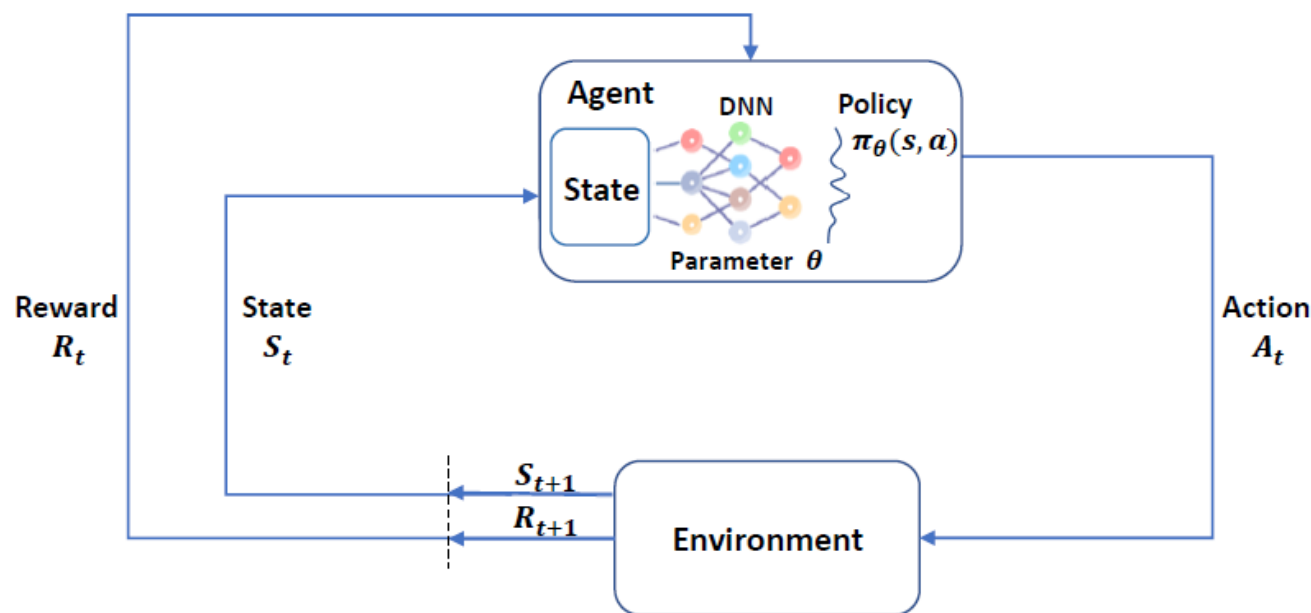
Reinforcement Learning Overview

- Reinforcement Learning (RL) is a learning paradigm where a single agent maximizes rewards by taking actions based on the environment's state and learning from the outcomes.



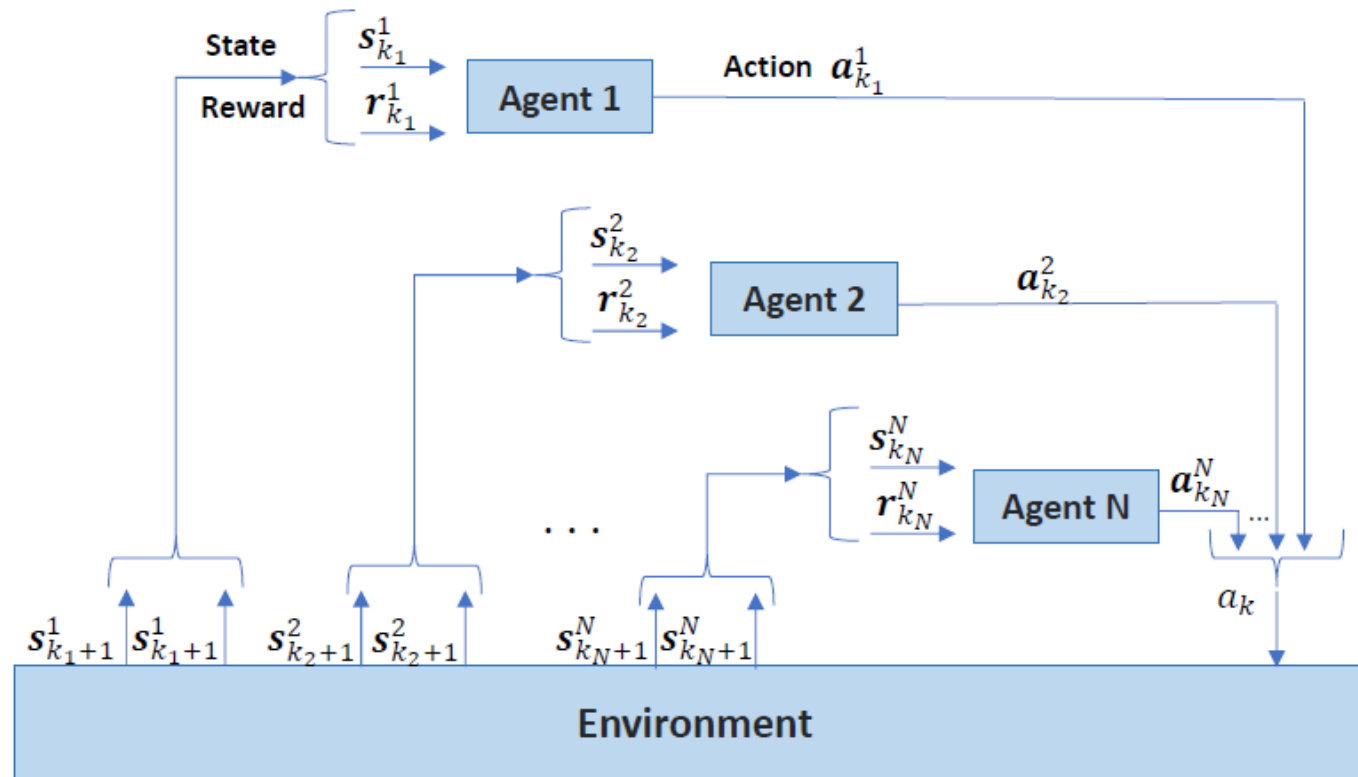
Deep Reinforcement Learning Overview

- Deep Reinforcement Learning (DRL) combines RL with deep neural networks (DNNs) to handle high-dimensional state spaces.



Multi-Agent Reinforcement Learning Overview

- Multi-Agent Reinforcement Learning (MARL) is where multiple self-learning agents share the same environment aiming to maximize their reward individually based on their local information.



Single-Agent RL Advantages and Disadvantages

➤ Advantages

- RL is ideal for complex optimization problems.
- RL has the ability to learn from experience using the state of the network and adapt to changing network conditions.
- Examples of RL algorithms include Deep Q Learning (DQN) and Deep Deterministic Policy Gradient (DDPG).

Single-Agent RL Advantages and Disadvantages

➤ Advantages

- DQN uses deep neural networks to approximate the action-value function in Q-learning, while DDPG learns deterministic policies for continuous action spaces as an actor-critic algorithm.
- These DRL algorithms have achieved success in domains like robotics, gaming, and optimization.
- By utilizing DQN and DDPG, we seek to optimize the collision avoidance mechanism in IEEE 802.11 standard.

Single-Agent RL Advantages and Disadvantages

➤ Disadvantages

- RL-based algorithms are complex, presenting high computational requirements.
- Require fine-tuned hyperparameters.
- The training process might take several hours and need extensive exploration.
- Significant amount of computing power.
- An occurrence of a sudden failure in a centralized single-agent RL leads to significant drop of the whole network operation.

Multi-Agent RL Advantages and Disadvantages

➤ Advantages

- Accelerated convergence to optimal network solutions in terms of performance.
- By leveraging distributed computing, new agents can be easily added to the corresponding scenario.
- Ensuring a high scalability level.
- Depending on the training mode and the objective, agents can exchange information, acquiring experience for increased convergence.
- The multi-agent system is intrinsically robust.

Multi-Agent RL Advantages and Disadvantages

➤ Disadvantages

- The non-stationarity problem leads to the problem of moving targets.
- The optimal policy will change whenever the policies of other agents change.
- The computational complexity grows exponentially as the number of agents being added increases.
- Making agents work collaboratively is a challenging task.

Proposed Centralized Solution

- In the centralized solution, the single-agent RL is located at the access point (AP). Since it has a global view of the network, a unique CW value is optimized and broadcast to all associated stations, ensuring that all stations have the same CW value.
- The AP chooses the best CW value that will be used across the associated stations to optimize the network's performance.

Proposed Mapping for Centralized Solution

- We propose deploying RL algorithms (DQN and DDPG) in a centralized manner to optimize collision avoidance in IEEE 802.11 networks. In order to do that, we used the following metrics:
 - **Observation:** uses two observation types, the *Average Transmission Queue Level*, which provides information about the network's congestion and packet backlog, and the *Collision Probability*, which informs about the transmission failure probability.
 - **Reward:** Network's overall throughput, which measures the amount of data successfully transmitted over a given time.
 - **Actions:** Contention window (CW) value, which determines the duration for which a station defers its transmission after detecting a collision.

Proposed Decentralized Solution

- The decentralized solution treats each station as an RL agent that updates its CW values resulting in optimized individual throughputs that are passed to the AP which sums up the individual throughputs from the self-learning agents.
- The AP broadcasts the total throughput value back to the stations maximizing the network's overall performance leading to a more flexible, adaptable and robust collision avoidance solution.

Proposed Mapping for the Decentralized Solution

- We propose deploying RL algorithms (DQN and DDPG) in the decentralized solution to optimize collision avoidance in IEEE 802.11 networks. In order to do that, we used the following metrics:
 - **Observation:** the Collision Probability, which informs about the transmission failure probability.
 - **Reward:** Network's overall throughput, which measures the amount of data successfully transmitted over a given time.
 - **Actions:** Contention window (CW) value, which determines the duration for which a station defers its transmission after detecting a collision.

Proposed Solution

- These metrics allow us to formulate the collision avoidance problem as a RL task.
- The RL agent learns to select optimal CW values based on the observed transmission queue level for the centralized case and uses the collision probability in the decentralized case to maximize the throughput.
- Through this two approach, we aim to enhance network performance by reducing collisions, and, consequently, increase the overall throughput.

Simulation Setup

- The NS-3 network simulator and the NS3-gym framework are used to create the agent's learning environment.
- The stations are deployed following static and dynamic scenarios with the AP learning and broadcasting the optimum CW value in the centralized case.
- The decentralized case the CW value is individually optimized for each station the resulting individual throughputs are summed up at the AP, the total throughput value is broadcast to the multiple stations.

Simulation Setup – Centralized Solution

- The stations are deployed following static and dynamic scenarios with the AP learning and broadcasting the optimum CW value in the centralized case.

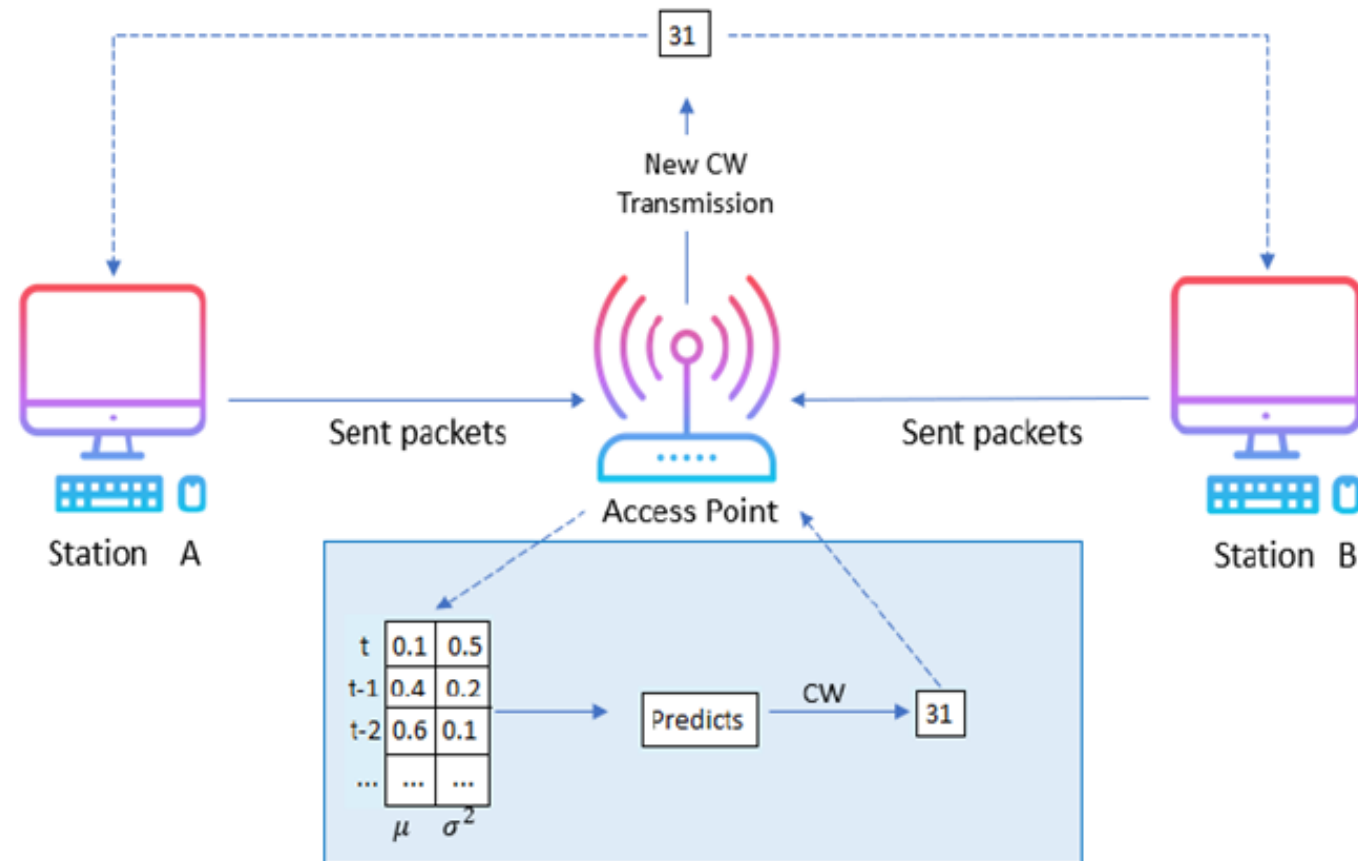


Figure 3.3: Scenario for assessing the proposed centralized DRL-based CW optimization method.

Simulation Setup – Decentralized Solution

- The decentralized case the CW value is individually optimized for each station the resulting individual throughputs are summed up at the AP, the total throughput value is broadcast to the multiple stations.

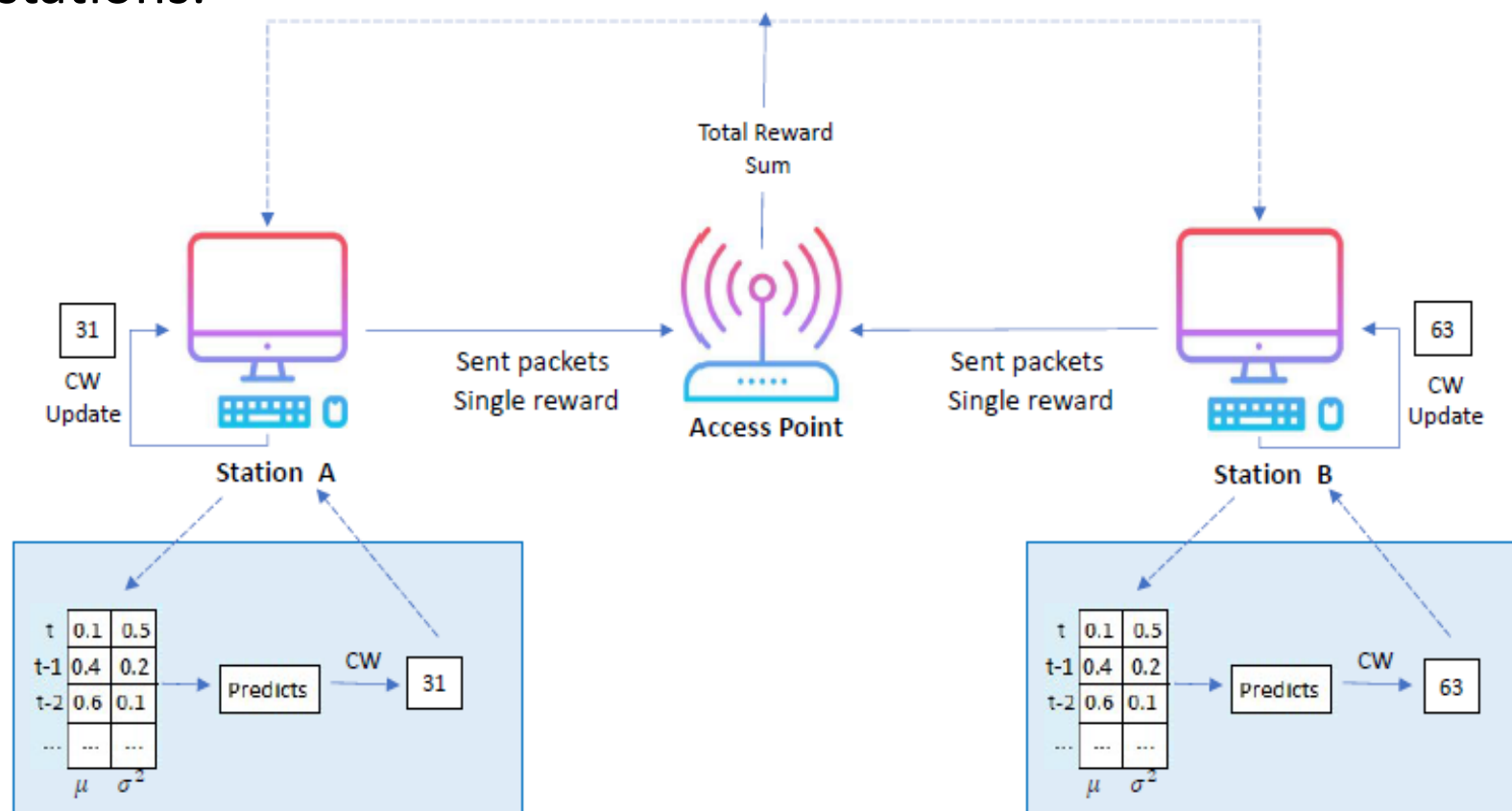


Figure 3.4: Scenario for assessing the decentralized DRL-based CW optimization method.

Simulation Setup

➤ Other simulation parameters are shown in the following tables.

Table 3.1: NS-3 Environment configuration parameters.

Configuration Parameter	Value
Wi-Fi standard	IEEE 802.11ax
Number of APs	1
Number of static stations	5,15, 30 or 50
Number of dynamic stations	increases steadily from 5 to 50
Frame aggregation	disabled
Packet size	1500 bytes
Max Queue Size	100 packets
Frequency	5 GHZ
Channel BW	20 MHz
Traffic	constant bit-rate UDP of 150 Mbps
MCS	HeMcs (1024-QAM with a 5/6 coding rate)
Guard Interval	800 [ns]
Propagation delay model	ConstantSpeedPropagationDelayModel
Propagation loss model	MatrixPropagationLossModel
Simulation time	60 [s]

Table 3.2: NS3-gym agent configuration parameters.

Configuration Parameter	Value
DQN's learning rate	4×10^{-4}
DDPG's actor learning rate	4×10^{-4}
DDPG's critic learning rate	4×10^{-3}
Reward discount rate	0.7
Batch size	32
Replay memory size	18,000
Size of observation history memory	300
<i>trainingPeriod</i>	840 [s]
<i>envStepTime</i> (i.e., interaction interval)	10 [ms]

Centralized Results for the Static Scenario

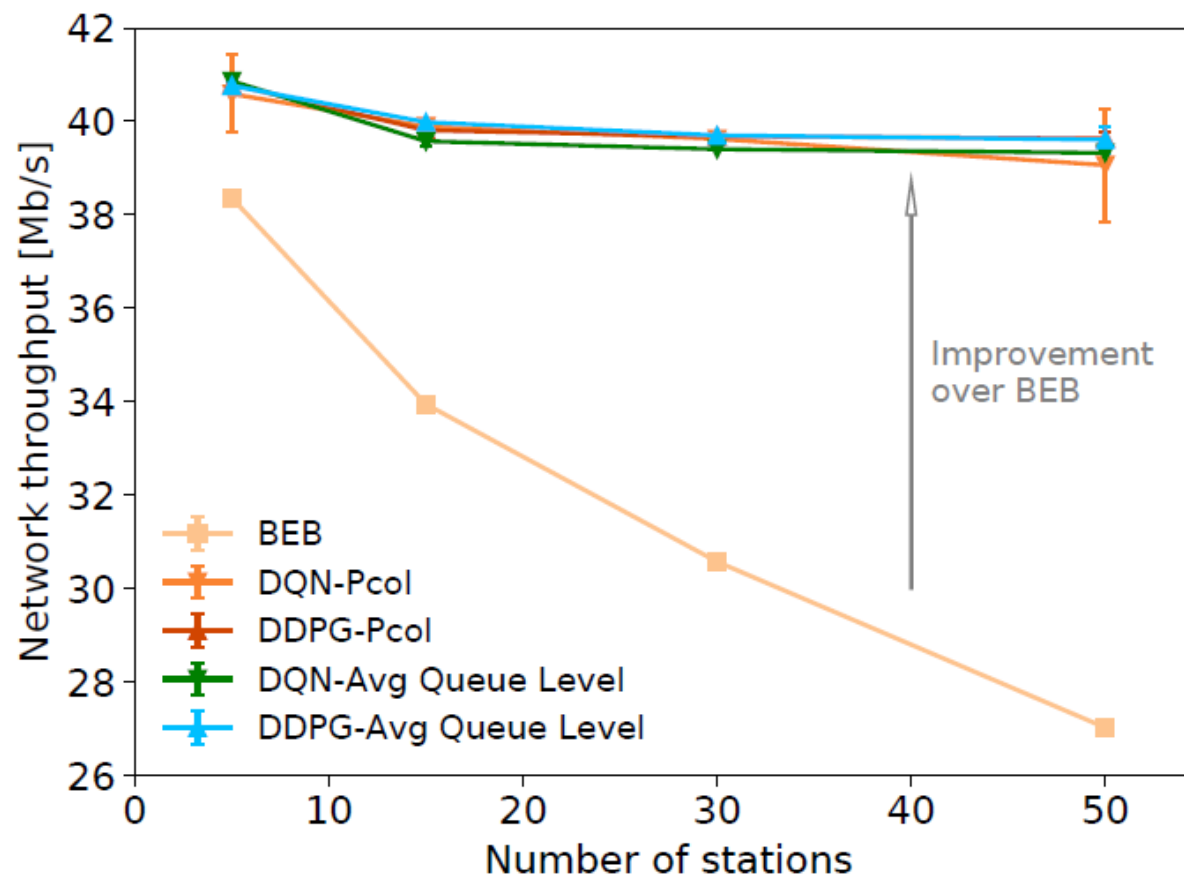


Figure 4.1: Comparison of the network throughput for the static centralized scenario as the number of stations increases from 5 to 50 using BEB, DQN, and DDPG with Pcol and Avg queue level as observations.

Centralized Results for the Static Scenario

➤ Mean CW value for different number of stations in a static centralized scenario using DQN and DDPG with Pcol as observation

Fig 4.2)
5 Stations

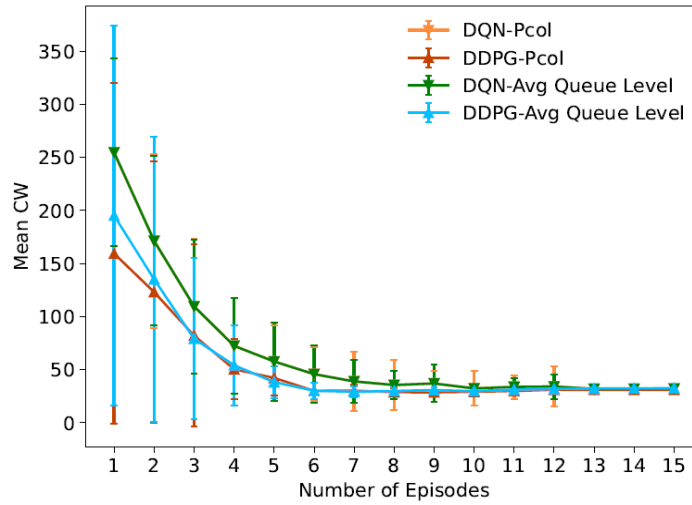


Fig 4.3)
15 Stations

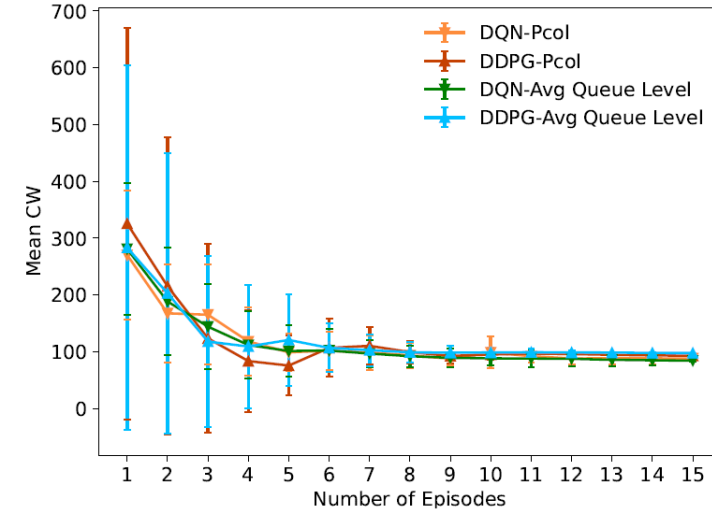


Fig 4.4)
30 Stations

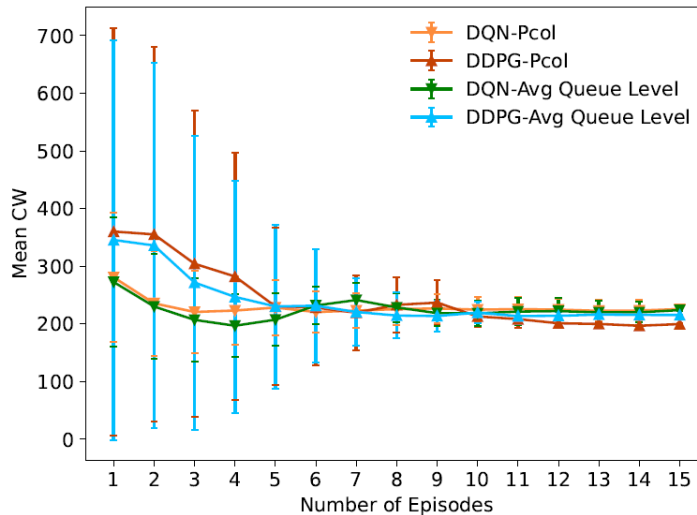
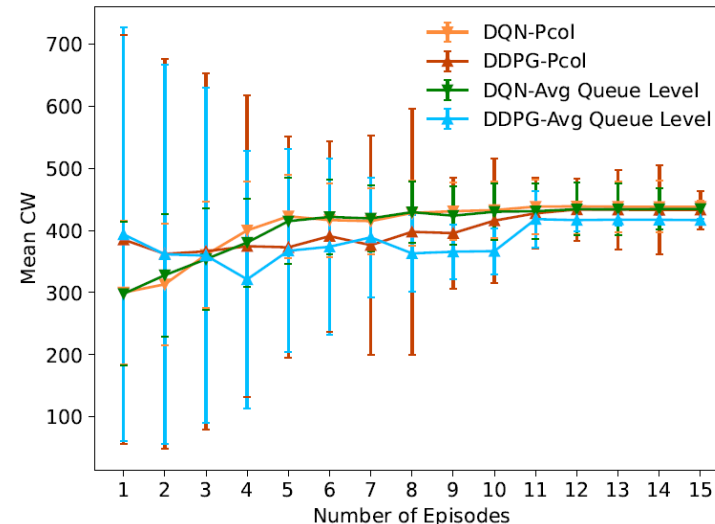


Fig 4.5)
50 Stations



Centralized Results for the Dynamic Scenario

➤ Mean CW value for different number of stations in a dynamic centralized scenario using DQN and DDPG with Pcol as observation

Fig 4.6)
5 Stations

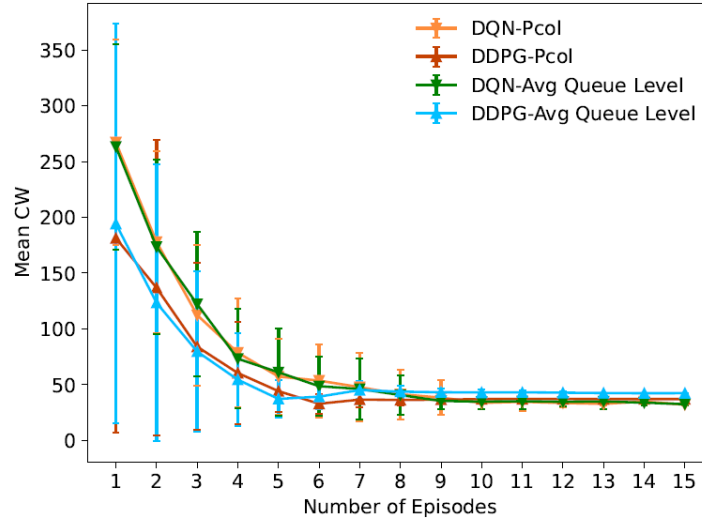


Fig 4.7)
15 Stations

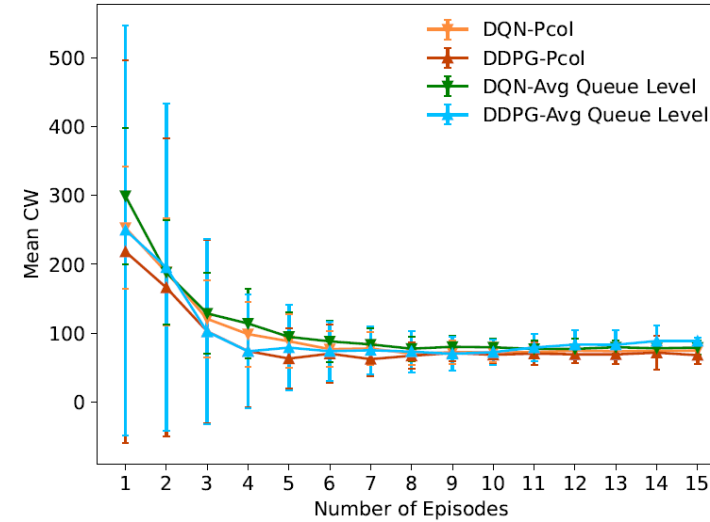


Fig 4.8)
30 Stations

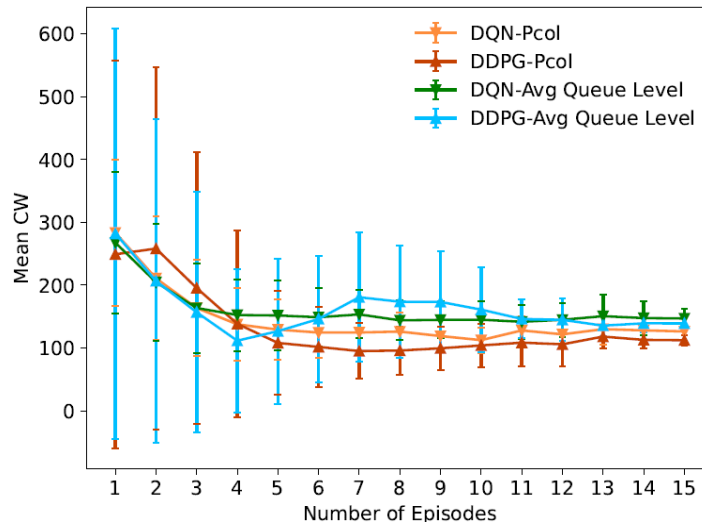
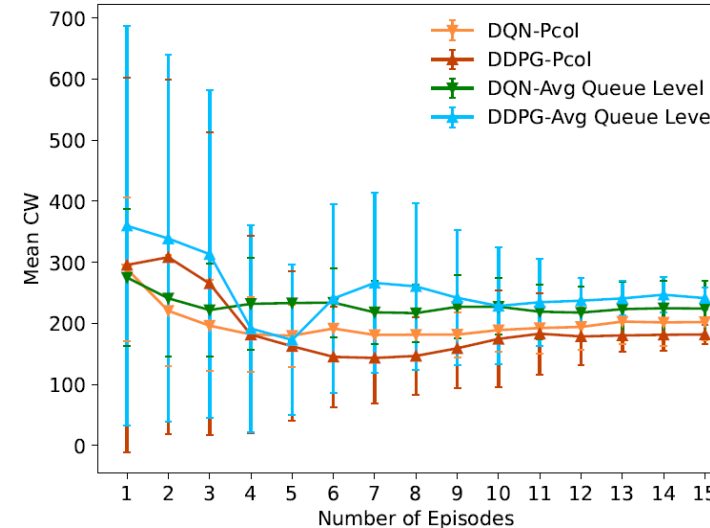


Fig 4.9)
50 Stations



Centralized Results for the Dynamic Scenario

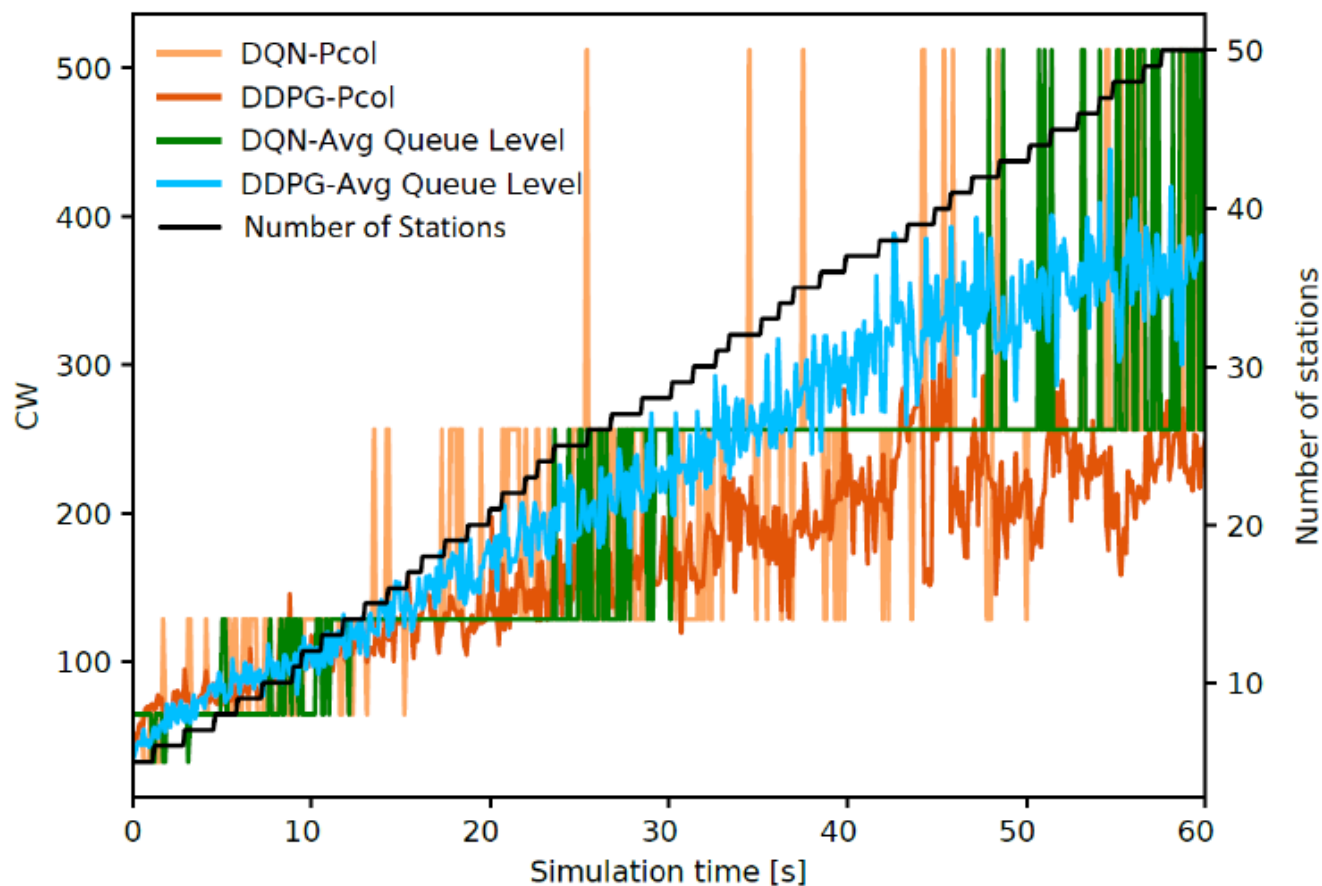


Figure 4.10: Selected CW value with the number of stations increasing from 5 to 50 in a dynamic centralized scenario using DQN, and DDPG with Pcol and Avg queue level as observations.

Centralized Results for the Dynamic Scenario

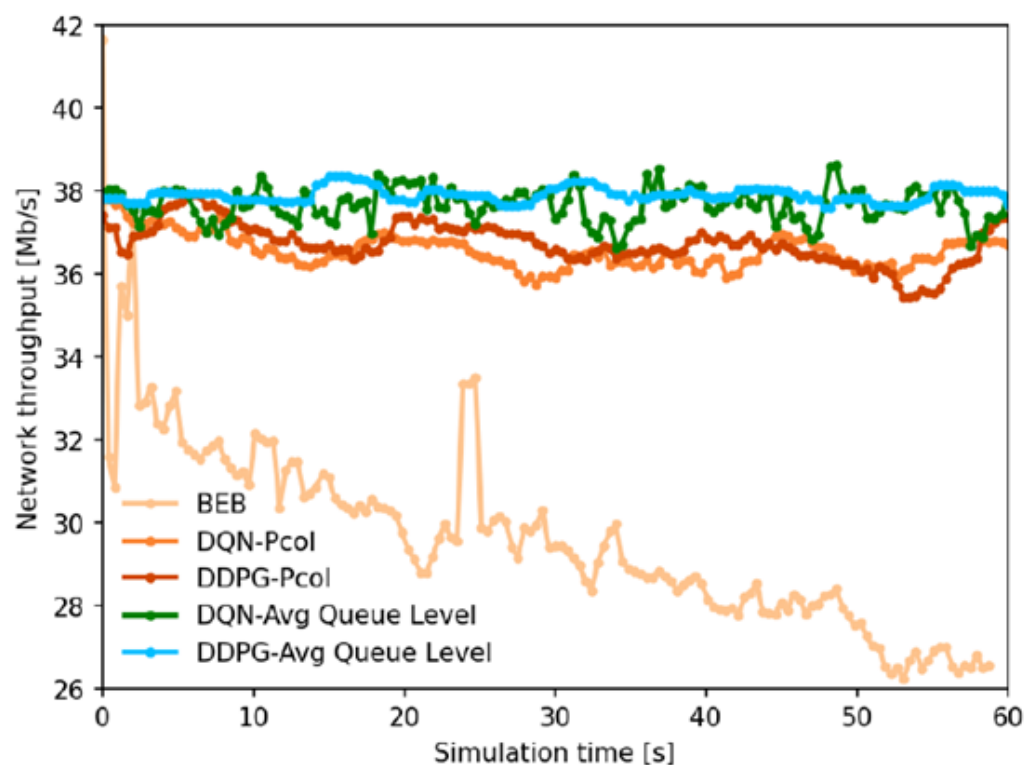


Figure 4.11: Comparison of the instantaneous network throughput in a dynamic centralized scenario using BEB, DQN, and DDPG with Pcol and Avg queue level as observations.

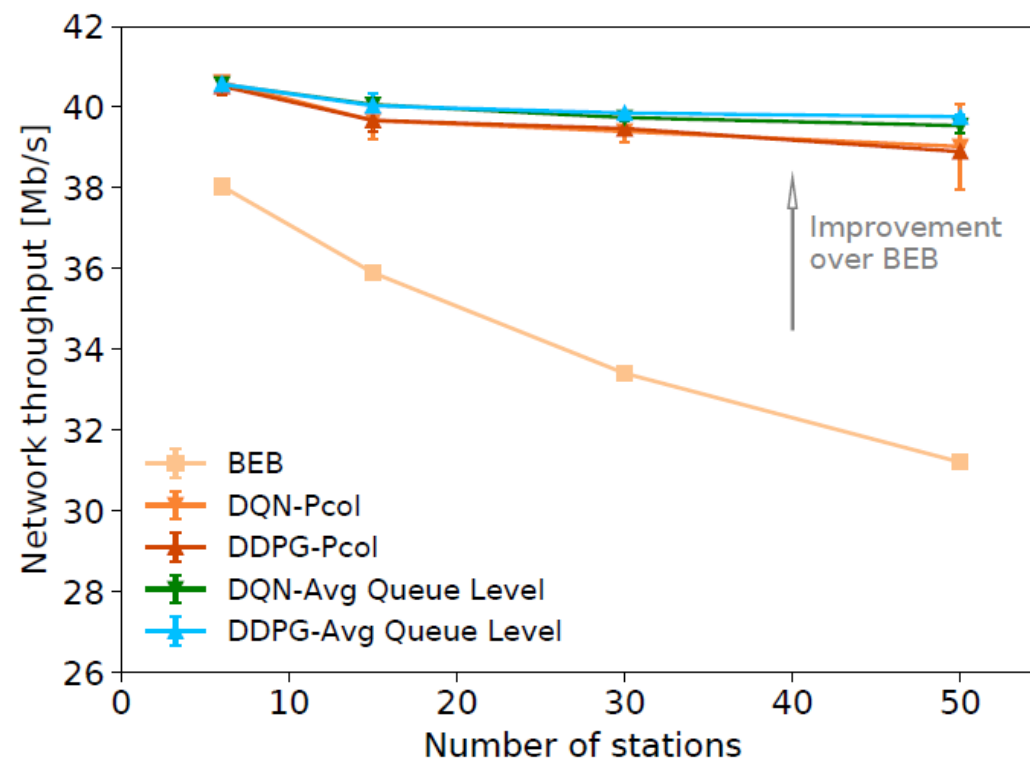


Figure 4.12: Comparison of the network throughput for the dynamic centralized scenario as the number of stations increases from 5 to 50 using BEB, DQN, and DDPG with Pcol and Avg queue level as observations.

Decentralized Solution Validation Results

Comparing the instantaneous network throughput

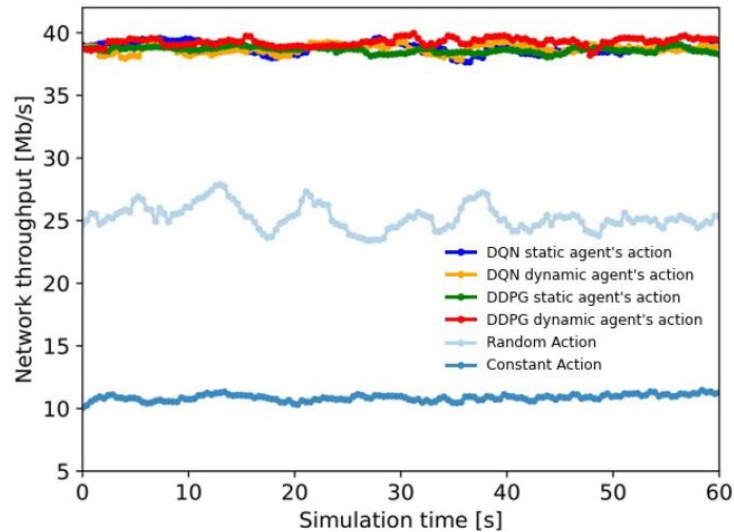


Fig 4.13)

Comparing the cumulative reward

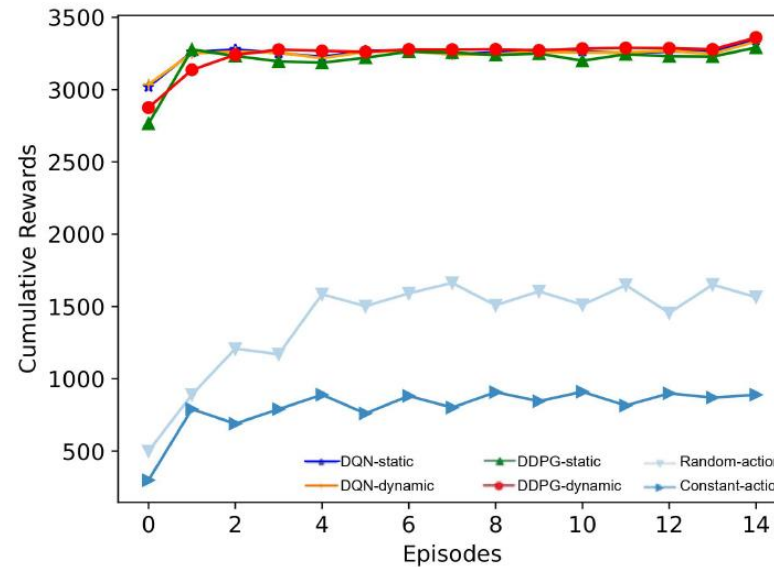


Fig 4.14)

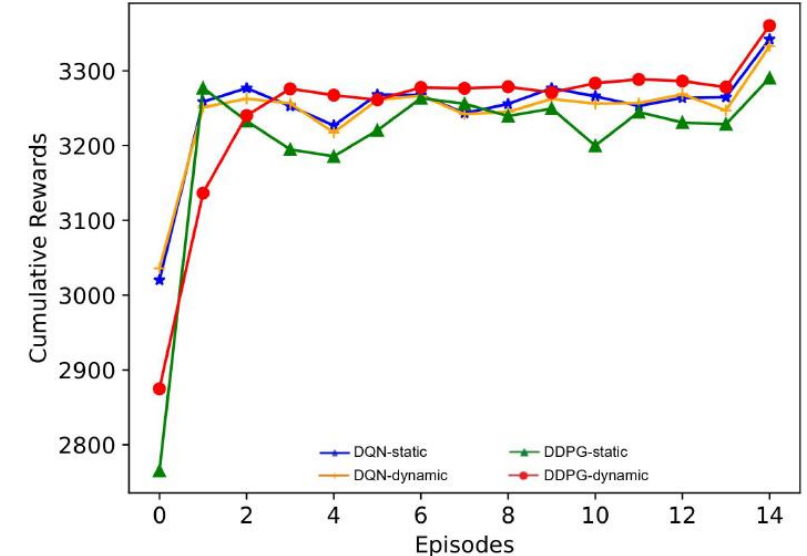


Fig 4.15)

Centralized vs Decentralized Results for the Static Scenario

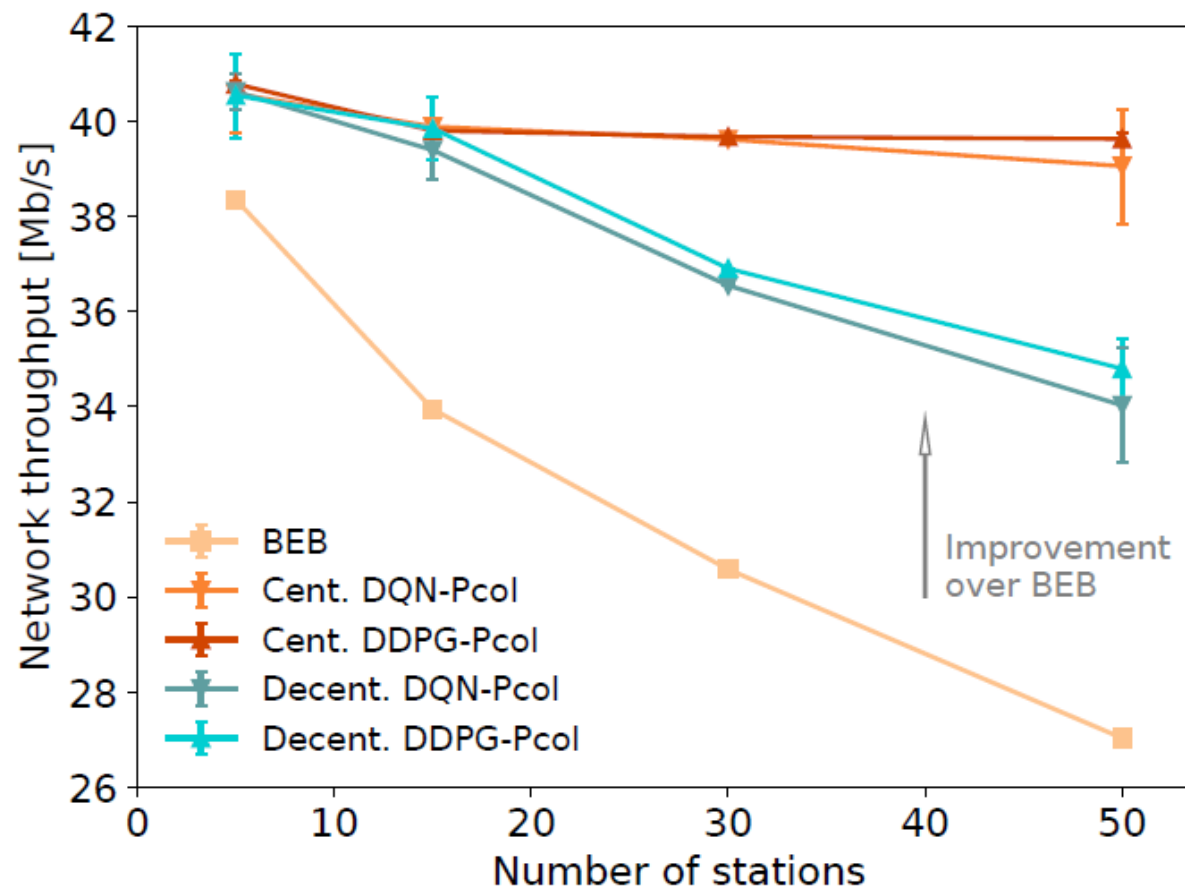


Figure 4.16: Comparison of the network throughput for the static centralized and decentralized scenario as the number of stations increases from 5 to 50 using BEB, DQN, and DDPG with Pcol as the observation.

Centralized vs Decentralized Results for the Static Scenario

➤ Mean CW value for different number of stations in a static Cent. and Decent. scenario using DQN and DDPG with Pcol as Obs.

Fig 4.17)
5 Stations

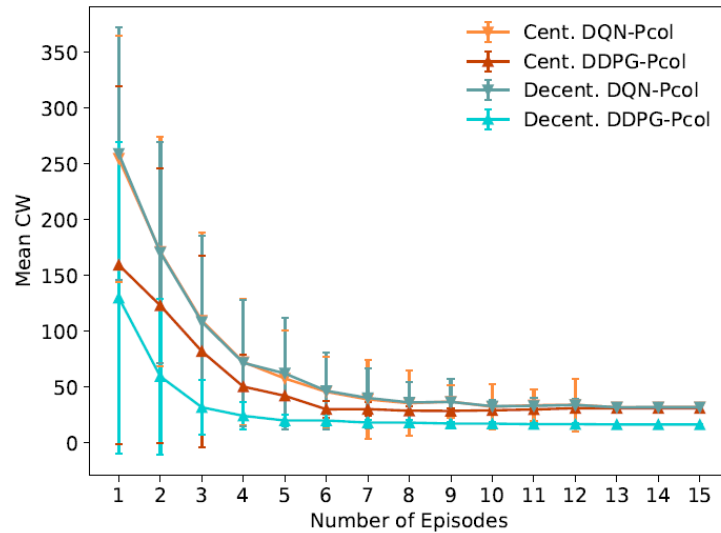


Fig 4.18)
15 Stations

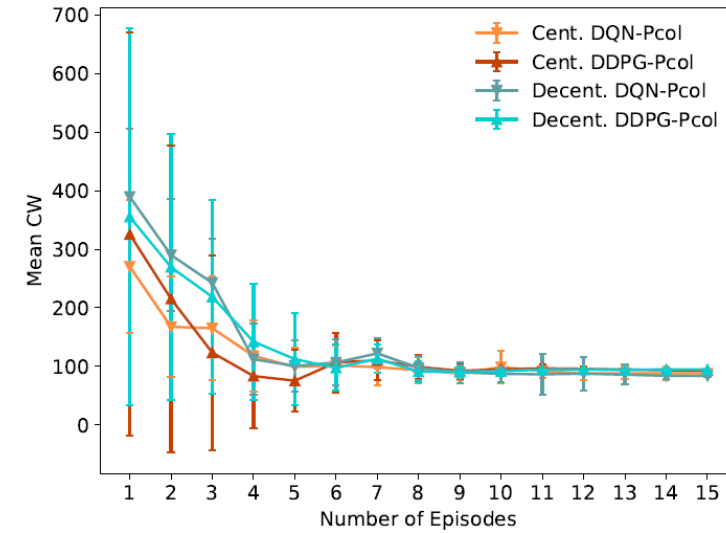


Fig 4.19)
30 Stations

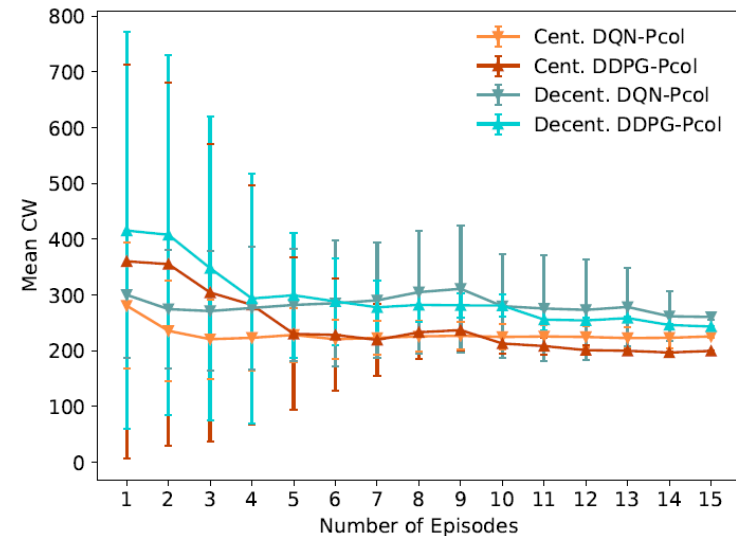
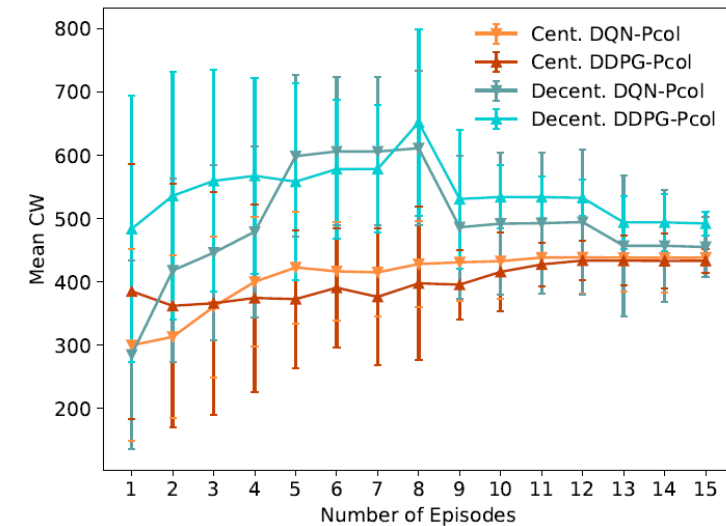


Fig 4.20)
50 Stations



Centralized vs Decentralized Results for the Dynamic Scenario

➤ Mean CW value for different number of stations in a dynamic Cent. and Decent. scenario using DQN and DDPG with Pcol as Obs.

Fig 4.21)
5 Stations

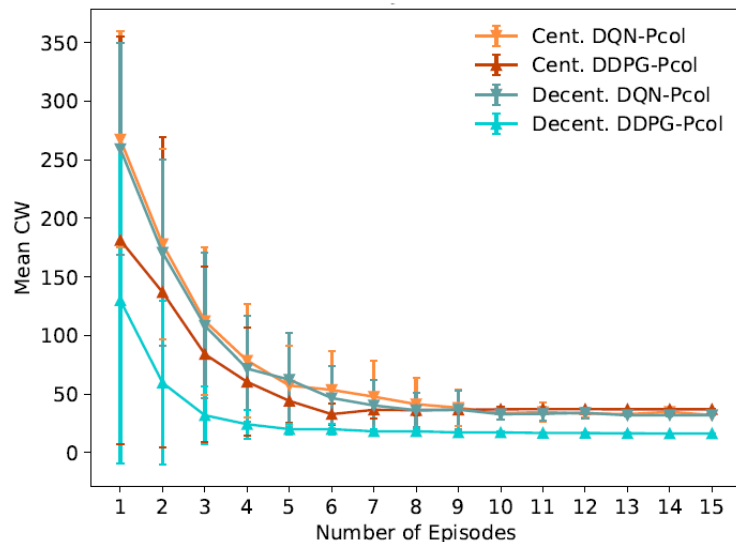


Fig 4.22)
15 Stations

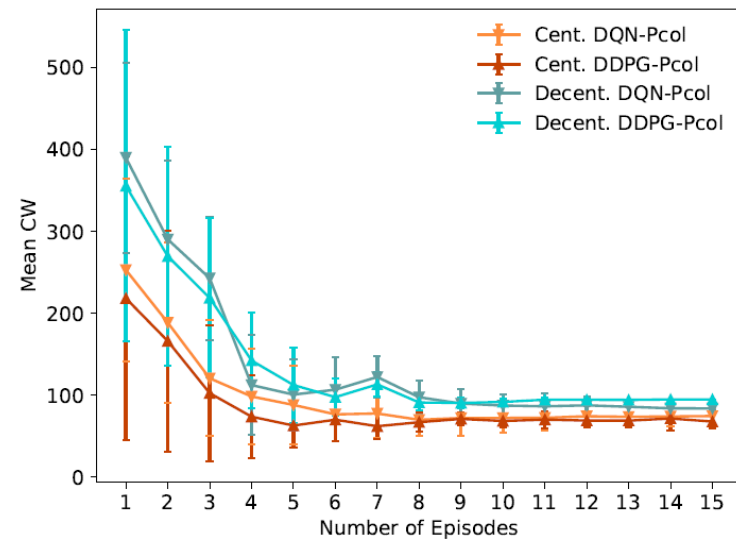


Fig 4.23)
30 Stations

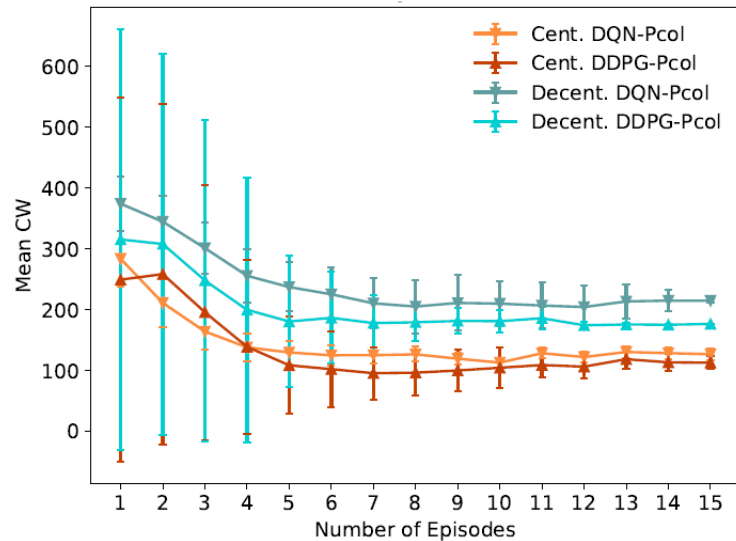
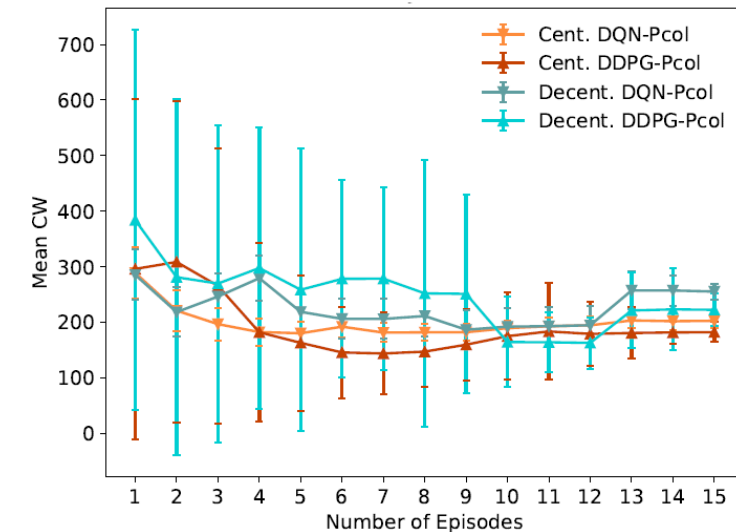


Fig 4.24)
50 Stations



Centralized vs Decentralized Results for the Dynamic Scenario

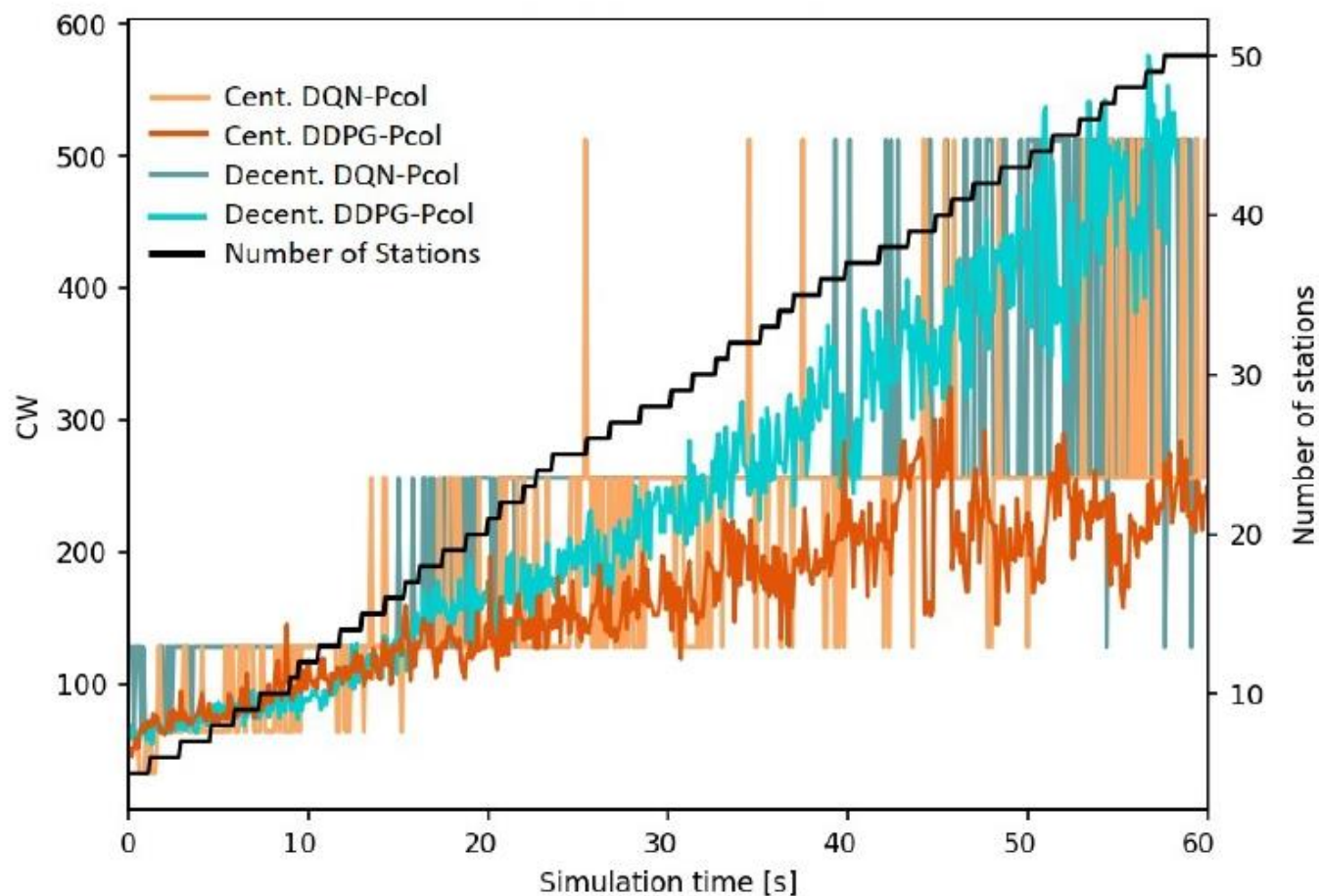


Figure 4.25: Selected CW value with the number of stations increasing from 5 to 50 in a dynamic centralized and decentralized scenario using DQN and DDPG with Pcol as observation.

Centralized vs Decentralized Results for the Dynamic Scenario

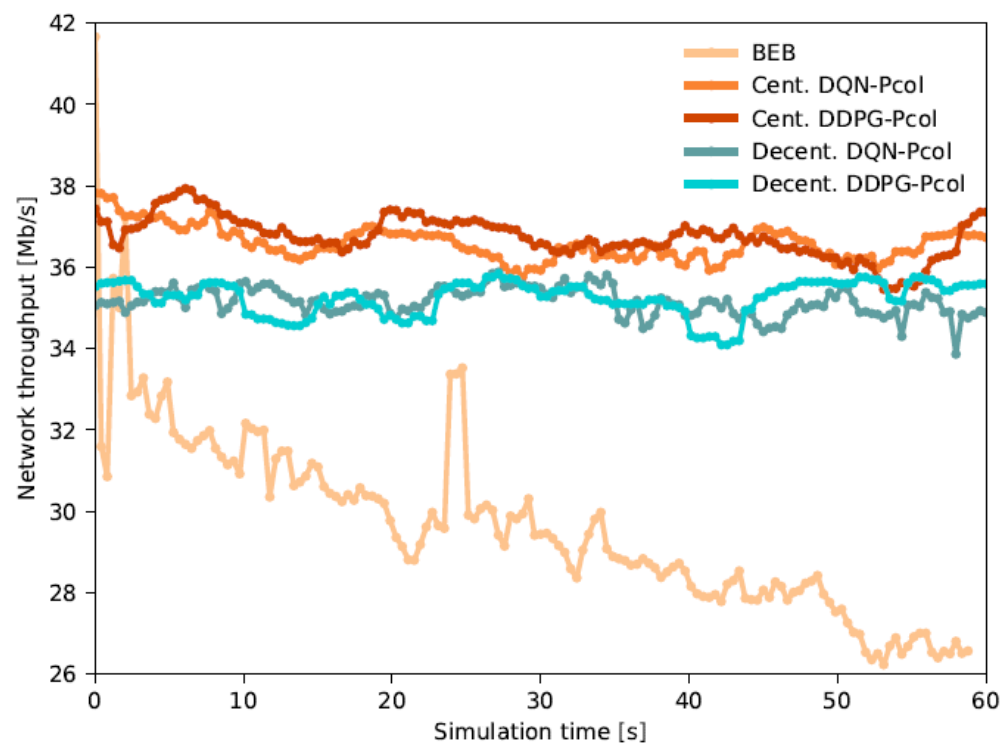


Figure 4.26: Comparison of the instantaneous network throughput in a dynamic centralized and decentralized scenario using BEB, DQN, and DDPG with Pcol as observation.

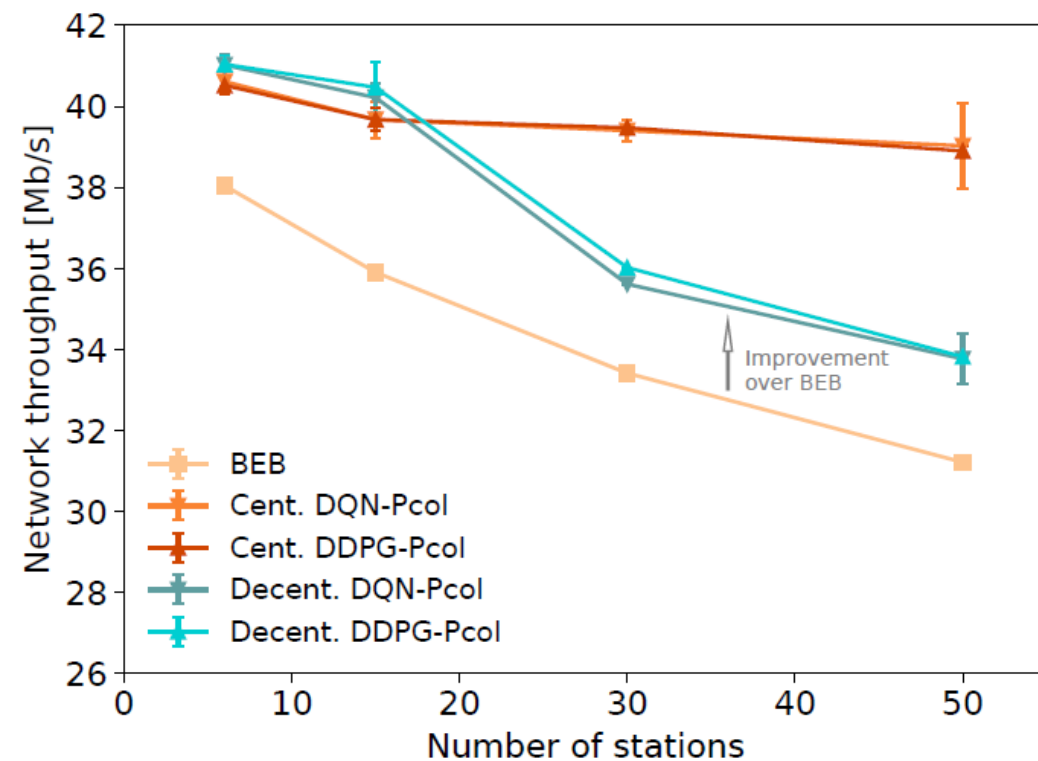


Figure 4.27: Comparison of the network throughput in a dynamic centralized and decentralized scenario as the number of stations increases from 5 to 50 using BEB, DQN, and DDPG with Pcol as observation.

Conclusions

- The thesis proposes the application of DRL algorithms (i.e., DQN and DDPG), as effective solutions to optimize the CW value. The proposed approach evaluates two observation metrics, namely the average of the normalized level of the transmission queue of associated stations and the collision probability (P_{col}).
- Two scenarios are considered: static and dynamic.
- Two approaches were considered: single-agent (centralized) and multi-agent (decentralized), highlighting the flexibility of DQN and DDPG in optimizing CW values under different network architectures.
- Simulation results show that the proposed solutions outperforms the 802.11 standard BEB algorithm by maintaining a stable throughput while reducing collisions for both scenarios.

Conclusions

- Our results show that the difference increases as the number of stations increases in the dynamic scenario, with DQN and DDPG showing a 45.52% increase in throughput with 50 stations compared to BEB in the centralized case, and for the decentralized case the degree of improvement reached up to 37.16%.
- Regarding the single versus multi-agent approach, the single-agent had a better performance because it has the ability to unilaterally make decisions that optimize the overall throughput in a given time. The multi-agent has a harder time coordinating what is best for the network because it is trying to maximize its own throughput.
- It can be concluded that either observation metric is suitable for obtaining the network's status and improving the optimal CW selection.
- It is concluded that all the proposed goals were, therefore, successfully achieved throughout the thesis.

Future Works

- Using other ML algorithms to optimize CW, such as Soft Actor-Critic (SAC) and Proximal Policy Optimization (PPO).
- Regarding the decentralized solution, future work could focus on making the stations cooperate by sharing information with each other.
- The assessment of the decentralized DRL-based solution using the averaged normalized transmission queues' level as the observation metric since it was proven by simulation that it provides a more accurate depiction of the network status.
- **Limitations:** The training process for the decentralized solution proposed in this work took quite a few hours and needed extensive exploration, requiring a considerable amount of computing power.

Thanks!

Santa Rita do Sapucaí, June 24th of 2024

Questions?

Santa Rita do Sapucaí, June 24th of 2024

Enhancing IEEE 802.11 Standard with Deep Reinforcement Learning for Optimal Channel Access

Sheila Cássia da Silva Janota Cruz

sheila.cassia@mtel.inatel.br

Advisor: Felipe Augusto Pereira de Figueiredo

Instituto Nacional de Telecomunicações - INATEL

Master in Telecommunications
Santa Rita do Sapucaí, June 24th of 2024